

Dice Rolling Experiment - Answer Key

Suggested responses for teachers

Year 11 Statistics

2026-08-08

Suggested Answers

Tip

These responses are intentionally flexible. Any reasonable explanation that recognises randomness and sample size should be accepted.

Part 1: Prediction

1. Any face could be predicted as most common, but students should recognise there is no guaranteed winner in a fair experiment.
2. No, the frequencies should not be exactly equal. Random variation means some faces will occur more often than others in a limited number of rolls.

Part 2: Record 20 Rolls

There is no single correct answer because the student data depends on the random outcomes.

Part 3: Frequency Table

For a fair die, each face has theoretical probability $\frac{1}{6}$, or about 0.167.

Part 4: Compare Results

1. The highest experimental frequency is whichever face has the largest count in the student data.
2. The lowest experimental frequency is whichever face has the smallest count in the student data.
3. The experimental probabilities are usually not exactly the same as the theoretical probabilities because the sample is finite and randomness causes variation.

Part 5: Reflection

As the number of rolls increases, the experimental probability should usually get closer to $\frac{1}{6}$, but it will still vary a little. This is an example of the law of large numbers.

Extension

With more rolls, the counts tend to become more stable and the bar chart usually looks more balanced, although some variation always remains.